OLIST E-COMMERCE / ANALYTICS PROJECT

Olist E-Commerce Analytics Report

Business performance • Data quality • Reconciliation • Recommendations

Executive focus

Turn nine related source tables into a reliable analytical model, then use consistent order, sales, payment, delivery and review measures to support decisions.

Documented control total	Value	Interpretation
Items + freight	R$ 15,843,553.24	Order-line value plus freight validation total
Payment value	R$ 16,008,872.12	Aggregate payment validation total
Gap to investigate	R$ 165,318.88	Payments exceed items + freight by approximately 1.04%

Dataset: Brazilian E-Commerce Public Dataset by Olist (2016–2018). The public dataset is described as containing approximately 100,000 orders and nine related CSV tables.

Prepared to address the supplied Final Project brief: define the business problem, model related tables, build KPIs and dashboards, and communicate evidence-based recommendations.

1. Executive Summary

This project analyzes the Olist Brazilian e-commerce dataset to connect commercial activity with customer location, products, sellers, payments, delivery dates and customer reviews. The model contains nine related tables and supports sales performance, operations, customer experience and financial-control questions.

The strongest documented quantitative finding is an aggregate reconciliation difference: the previously recorded items-plus-freight total is R$ 15,843,553.24, while the payment total is R$ 16,008,872.12. Payments are higher by R$ 165,318.88, approximately 1.04% of the items-plus-freight total. This is a control exception to investigate, not proof by itself of overpayment, revenue leakage or a loss.

Decision statement

Do not treat the model as financially reconciled until the difference is decomposed by order and explained. Preserve a transparent audit trail for every imputed or corrected source value.

Business problem

The business needs one consistent view of order volume, item sales, freight, payment value, geographic demand, delivery reliability and customer satisfaction. These facts live at different grains: one order can contain multiple items, multiple payment records and review records. Poorly aligned calculations can overcount orders or make totals appear inconsistent.

Analytical questions

- How do orders and sales value move over time?
- Which product categories and geographic markets contribute most to sales and order volume?
- How are payment value and payment methods distributed?
- How long does delivery take, and how often do deliveries miss the estimated date?
- How are customer review scores distributed, and where are low scores concentrated?
- Which orders explain the difference between item-plus-freight value and payments?

2. Dataset and Analytical Model

The source contains nine related CSV tables covering transactions and supporting dimensions. In this dataset, customer_id identifies the customer record associated with an order, while customer_unique_id is used to recognize repeat purchases by the same person. These identifiers should not be conflated.

Table	Analytical role	Key grain / identifier
olist_orders_dataset	Order lifecycle / delivery	One row per order; order_id
olist_order_items_dataset	Product lines, prices, freight	One row per order line; (order_id, order_item_id)
olist_order_payments_dataset	Payment value and method	One row per payment sequence; (order_id, payment_sequential)
olist_order_reviews_dataset	Customer feedback	Review records; review_id and order_id
olist_customers_dataset	Customer and delivery geography	customer_id; customer_unique_id for repeat-customer analysis
olist_products_dataset	Product attributes and category	product_id
olist_sellers_dataset	Seller and origin geography	seller_id
product_category_name_translation	Portuguese-to-English category label	product_category_name
olist_geolocation_dataset	ZIP-prefix coordinates	geolocation_zip_code_prefix; repeated in raw source

Modeling principles

- Count orders at order grain (orders.order_id), not by counting item rows or payment rows.
- Calculate sales and freight from the order-items table; calculate payment value from order payments.
- Use customer_unique_id for unique/repeat-customer metrics and customer_id for the order-to-customer record relationship.
- Use a proper Calendar table for consistent time intelligence; confirm its active relationship matches the date role used by each measure.
- Aggregate or validate geolocation at ZIP-prefix grain before using it as a one-side lookup dimension.

3. Data Preparation and Quality Assessment

The documented Power Query steps include type conversion, text trimming/cleaning, date handling, category translation, geolocation deduplication and derived delivery/reconciliation fields. Several transformations need explicit governance because they may affect quantitative conclusions.

Observed transformation	Risk / impact	Recommended control
Negative price values converted to absolute values	Changes source meaning; may turn an error or credit into a positive sale.	Keep raw value, flag changed rows, verify against source before revenue use.
Null price values replaced with 145.95	Fixed imputation can bias sales and average-price metrics.	Add an imputation flag; trace affected item/order; compare sensitivity with and without imputation.
Geolocation deduplicated on ZIP prefix only	Keeps one coordinate per prefix; spatial location may be arbitrary.	Aggregate coordinates (e.g., mean latitude/longitude by ZIP prefix) or document representative-point logic.
City placeholders replaced with city names	Broad text replacement can misclassify rows if original encoding errors vary.	Profile affected rows, verify mapping, and preserve original text in a staging column.
Order timestamps split to date fields	Time-of-day information is lost in final date-only columns.	Retain raw datetime columns if hourly or partial-day delivery duration matters.
Payment reconciliation performed after grouping	Order-level differences may be hidden by aggregate totals.	Retain order-level Difference and Reconciliation Status; investigate exceptions by order_id.

Quality gates before sign-off

- Record row counts before and after each cleaning operation.
- Check uniqueness of all expected keys after refresh; verify customer_id is unique in customers and order_id is unique in orders.
- Check blank keys, duplicate keys, orphan foreign keys and invalid date sequence flags.
- Confirm financial totals with independent sums and a row-level reconciliation sample.
- Document assumptions and imputation counts in the final project appendix.

4. Quantitative Findings: Financial Reconciliation

The following two totals are the project's documented validation checkpoints from prior analysis work. They are reproduced here as recorded values; they were not recomputed independently from raw CSVs during preparation of this report.

Control metric	Documented value	How to interpret
Items + freight	R$ 15,843,553.24	Sum of item prices plus freight values
Payment value	R$ 16,008,872.12	Sum of payment_value across payment records

Difference (payments - items + freight)	R$ 165,318.88	Payment total is higher
Difference / items + freight	1.04%	Size of the gap relative to items + freight

What this does and does not mean

- It establishes an aggregate reconciliation exception that deserves investigation.
- It does not establish that every order is wrong; positive and negative order-level differences may offset or accumulate.
- It does not prove that the gap is revenue, profit, loss, refund exposure or a payment error.
- Potential explanations to test include multiple payment rows, cancelled/unavailable orders, incomplete item coverage, refunds/adjustments, cleaning imputation and source-level differences. These are hypotheses, not confirmed causes.

Recommended next calculation

Group both item-plus-freight and payment_value by order_id, join the two summaries, calculate per-order Difference and Reconciliation Status, then rank the largest absolute differences and inspect a sample from each status.

5. Dashboard and KPI Framework

The interactive Power BI report should support two complementary views: an overview dashboard and a deeper operational/customer-experience analysis. Use the same definitions in Excel, Power BI, SQL and the report.

Dashboard view	Core KPIs / visuals	Decision supported
Overview	Total Orders; Item Sales; Freight; Payment Value; Unique Customers; Average Review Score; orders over time; sales by category; orders by state; payment mix.	Understand scale, trend, commercial mix and geographic demand.
Delivery & Customer Experience	Average Delivery Days; On-Time Delivery Rate; Late Orders; Average Delay Days; review score distribution; delivery quality flags; reconciliation exceptions.	Identify operational bottlenecks, customer pain points and orders needing investigation.

Measure definitions to standardize

Metric	Recommended definition / warning
Total Orders	Distinct count of olist_orders_dataset[order_id]. Do not count item or payment records.
Item Sales	SUM of order_items[price]. Keep freight separate unless reporting explicitly labels sales including freight.
Freight	SUM of order_items[freight_value].
Payment Value	SUM of order_payments[payment_value].
Unique Customers	Distinct count of customers[customer_unique_id], not customer_id.
Delivery Days	Difference between delivered-customer date and purchase date; explicitly state calendar-day vs elapsed-time logic.
On-Time Delivery Rate	Delivered orders whose delivered date is on or before estimated date divided by eligible delivered orders; exclude missing dates and disclose denominator.
Average Review Score	AVERAGE of review_score, with review grain and missing review coverage disclosed.

6. Insights, Recommendations and Action Plan

Evidence-backed observations

- Financial reconciliation: documented payment total exceeds documented items-plus-freight total by R$ 165,318.88 (approximately 1.04%). Investigate order-level exceptions before accepting the totals as reconciled.
- Data correction risk: converting negative prices to absolute values and imputing null prices to 145.95 can alter revenue measures unless changed/imputed rows are flagged and justified.
- Geographic precision risk: deduplicating geolocation solely on ZIP prefix can preserve an arbitrary coordinate instead of a representative location.
- Metric grain risk: order, order-item, payment and review tables have different grains; mixing raw numeric columns in one visual can duplicate orders or inflate sums.
- Identity distinction: customer_id represents the order-specific customer record in this dataset, while customer_unique_id supports repeat-customer analysis.

Prioritized recommendations

Priority	Action	Expected outcome
P1 - Financial control	Reconcile at order_id; investigate top absolute gaps and document accepted explanations.	Transparent, auditable payment-to-order control.
P1 - Data quality	Retain raw values and add flags for negative-price conversion, null-price imputation and invalid date sequences.	Prevent silent distortion of revenue and delivery KPIs.
P1 - Semantic model	Keep measures aligned to table grain; use distinct order counts; test relationship cardinality and filter direction.	Avoid duplicated order counts and double-counted amounts.
P2 - Geography	Aggregate ZIP geolocation deterministically; validate city/state mappings and encoding repairs.	More reliable map and state/city analysis.
P2 - Delivery	Track on-time rate, median/average delivery days, delay days and missing/invalid dates by month and state.	Prioritize operational improvement based on comparable measures.
P3 - Customer experience	Segment review scores by category, delivery performance and time; read low-score comments only with appropriate privacy handling.	Target service improvements linked to customer feedback.

7. Limitations, Reproducibility and References

Limitations

- Detailed segment rankings (top categories, states, cities, payment methods and low-score product groups) are not stated as numerical findings here because validated result tables were not available for independent recomputation at report-authoring time. These should be quoted from refreshed Power BI visuals or reproducible query output.
- The aggregate validation figures are carried forward from project work and should be independently recalculated from the final refreshed model before submission.
- The null-price replacement value and negative-price correction require explicit justification; if they were only temporary repair steps, update the report after reviewing affected rows.
- The dataset describes historical Brazilian marketplace transactions from 2016–2018. Results should not be interpreted as current market performance.

Reproducibility checklist

- Retain source CSVs and cleaning logic; record file origin and refresh date.
- Record table row counts and key uniqueness results after refresh.
- Save final DAX measure definitions, relationship diagram and dashboard screenshots.
- Use the same measure definitions across Python/SQL/Excel/Power BI, as required by the project brief.
- Keep an order-level exception extract as evidence for reconciliation conclusions.

References

[1] Olist. (2018). Brazilian E-Commerce Public Dataset by Olist. Kaggle. https://www.kaggle.com/olistbr/brazilian-ecommerce.